MILITARY SPECIFICATION

ADHESIVE, AIR-DRYING, SILICONE RUBBER

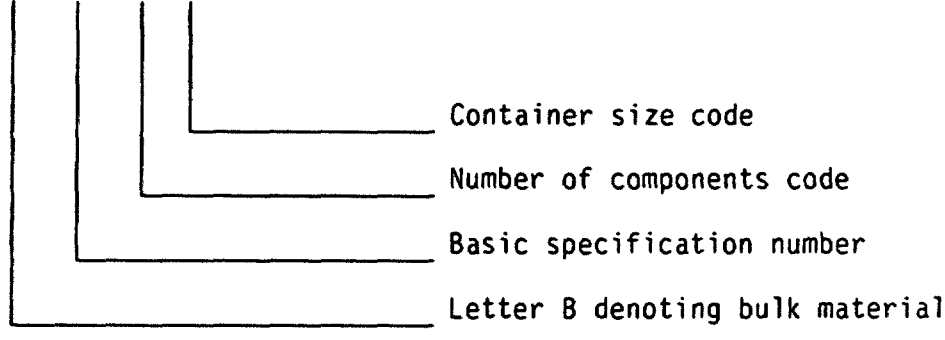
This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic adhesive suitable for use in bonding silicone rubber to itself and to aluminum without the use of heat and pressure. The product may be either a one-part or a two-part adhesive.

1.2 Military part number. The military part number shall consist of the letter B, plus the basic number of this specification, followed by a number indicating either one or two components, and a set of two letters specifying container size (see table I for component and size codes).

Example: B 47317 1 QT



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document/Improvement Proposal (DD Form 1426) at the end of this document or by letter.

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TABLE I. Military part number codes

Number of components	Code	Container size 1/	Code
1----------	1	Five ounces	OZ
2----------	2	One pint	PT
		One quart	QT
		One gallon	GL

1/ Wherever inch pound units of measurement are expressed in this specification, metric equivalents shall be acceptable.

For two-component material, this table indicates the total size that will result from mixing according to the manufacturer's instructions.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-250/5 Aluminum Alloy, Alclad 2024, Plate and Sheet

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

MIL-STD-45662 Calibration System Equipment

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D413	Rubber Property, Adhesion to Flexible Substrate, Standard Test Method for
ASTM D471	Rubber Property, Effect of Liquids, Standard Test Method for
ASTM D1084	Viscosity of Adhesives, Standard Test Method for
ASTM D1810	Specific Gravity of Ashorels, Standard Test Method for
ASTM D3951	Commercial Packaging, Standard Practice for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Materials. The adhesive covered by this specification shall be a silicone rubber material which will produce a product meeting all the requirements specified herein (see 4.5 and 4.6).

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3.3 Curing time. The material shall fully cure within 72 hours at room temperature of 23 ± 2° Celsius(C) or 73 ± 4° Fahrenheit(F). See 4.6.2 and table II.

3.4 Pot life. Pot life shall be not less than three hours at room temperature (see 4.6.3).

3.5 Viscosity. Viscosity shall be 1000 to 5000 centipoises (see 4.6.4).

3.6 Specific gravity. Specific gravity shall be 0.984 to 1.008 (see 4.6.5).

3.7 Peel strength. Peel strength shall be not less than indicated in table II (see 4.6.6).

TABLE II. Peel strength

Type of bond	Peel strength	
	24-hour cure	72-hour cure
Silicone rubber to silicone rubber	8 pounds/inch	10 pounds/inch
Silicone rubber to aluminum	8 pounds/inch	10 pounds/inch

3.8 Water resistance. When peel tested after 24-hour immersion in water as specified in 4.6.7, the adhesive properties of silicone rubber to clad aluminum shall not have decreased more than 15 percent of the 72 hour cure adhesion value.

3.9 Oil resistance. When peel tested after immersion in oil as specified in 4.6.8, the adhesive properties of silicone rubber to aluminum shall not have decreased more than 20 percent of the 72 hour cure adhesion value.

3.10 Shelf life. The adhesive (and catalyst, if a two-part product) shall not gel and shall meet the requirements of this specification after being subjected, in the original unopened container, to a four week storage period at 50 ± 2°C (122 ± 4°F) (see 4.6.9).

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3.11 Marking for identification. 1/ Marking for identification shall be in accordance with MIL-STD-130 and shall include, but not be limited to, the following (see 4.6.1).

a. The number, date, and title of this specification.

b. The military part number.

c. Identification of each component (base or catalyst if a two-component adhesive, or "single component").

d. Manufacturer's name or trademark.

e. Date of manufacture.

f. Any applicable warnings.

g. Complete mixing instructions for two-component material.

3.12 Workmanship. The material shall be smooth, homogeneous, and free from lumps or any foreign matter which might prevent the material from meeting the requirements of this specification (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with

1/ Where unit container size precludes complete marking, such marking shall appear on the intermediate container, except that component identification (single component, base, or catalyst) shall appear on the unit container.

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all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection and testing of the adhesive shall be classified as:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under ambient conditions.

4.3.1 Inspection equipment accuracy. Equipment used to measure test conditions and test parameters shall be calibrated in accordance with MIL-STD-45662. Where the state of the art permits, basic precision equipment capable of measuring or testing to an accuracy representing one-tenth of the prescribed tolerance on the inspection equipment shall be used.

4.4 First article inspection.

4.4.1 First article sample. When specified in the contract or purchase order (see 6.2.1), a first article sample of a quantity of adhesive sufficient to perform all tests specified herein shall be provided. The first article sample shall be fabricated using the same processes, facilities, and materials as are proposed for production. The first article shall meet all requirements of this specification.

4.4.2 Action in case of failure. No failures shall be permitted during first article sample inspection. If a failure occurs, the first article sample shall be rejected. No units shall be offered for acceptance until approval of a first article sample has been obtained from the procuring activity.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the tests as specified herein.

4.5.1 Lot size. Lot size shall consist of all the adhesive manufactured in one production batch run from one process forming part of one contract or order, and submitted for acceptance at one time.

4.5.2 Sampling. Unless otherwise specified, sampling shall be performed on a quantity of adhesive sufficient to assure that all requirements of this specification have been met.

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4.5.2.1 Action in case of failure. If a sample fails to pass all inspections, acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring activity has been taken. After corrective action has been taken, inspections shall be repeated on an additional sample unit.

4.6 Test methods.

4.6.1 Visual inspection. Visual inspection shall verify compliance with 3.2, 3.11, 3.12, 5.1, 5.2, and 5.3.

4.6.2 Curing time. Successful completion of the peel strength tests shall verify compliance with 3.3.

4.6.3 Pot life. Spread 10 grams of single component or mixed adhesive onto a non-porous surface. Maintain room temperature. If the material remains spreadable after three hours, the requirement of 3.4 has been met.

4.6.4 Viscosity. Viscosity shall be determined in accordance with ASTM D1084. Verify compliance with 3.5.

4.6.5 Specific gravity. Specific gravity shall be determined in accordance with in ASTM D1810. Verify compliance with 3.6.

4.6.6 Peel strength.

4.6.6.1 Preparation of specimens. Rubber-to-rubber specimens shall be made by bonding 1.25 x 6 inch strips of glass fabric reinforced by silicone rubber tape prepared in accordance with the manufacturer's recommendations. The strips shall have an overall thickness of 0.045 to 0.050 inches and shall be cured for ten minutes at 150°C (302°F). These strips shall have a post cure of one hour at 150°C (302°F) plus four hours at 250°C (482°F). Rubber-to-aluminum specimens shall be made by bonding a 1.5 x 6 inch strip of the tape specified above to 2 x 6 inch strip of clad aluminum conforming to QQ-A-250/5. Surface preparation of the adherends shall be in accordance with the manufacturer's recommendation.

4.6.6.2 Test procedure. The test shall be conducted in accordance with ASTM D413. All peel tests shall be run at standard conditions (see 4.3) and four specimens of each bond being tested shall be used for each peel test being run.

4.6.6.3 Adhesion. Rubber-to-rubber and rubber-to-aluminum specimens, prepared in accordance with 4.6.6.1, shall be cured at room temperature for the periods of time specified in table II. The cured specimens shall then be peel tested as specified in 4.6.6.2. Verify compliance with 3.7.

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4.6.7 Water resistance. Test specimens, prepared in accordance with 4.6.6.1, shall be air cured at room temperature for 72 hours after which they shall be immersed in water at room temperature for 24 hours. The specimens shall then be peel tested as specified in 4.6.6 within five minutes after removal from the water. Verify compliance with 3.8.

4.6.8 Oil resistance. Test specimens, prepared in accordance with 4.6.6.1, shall be air cured for 72 hours at room temperature and then immersed in oil conforming to ASTM D471 for 24 hours at room temperature. The specimens shall then be peel tested as specified in 4.6.6 within five minutes after removal from the oil. Verify compliance with 3.9.

4.6.9 Shelf life. One unopened quart container of adhesive (and necessary unopened catalyst if a two-part product) shall be placed in an oven maintained at a temperature of 50° + 2°C (122° + 4°F) and all allowed to age for four weeks. It shall then be allowed to cool at room temperature for four hours and examined for signs of gelling or deterioration other than partial solvent loss, assuring compliance with 3.10.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with ASTM D3951. Adhesive shall be packaged to afford adequate protection against deterioration and damage in accordance with supplier's commercial practice (see 4.6.1).

5.2 Packing. Packing shall be in accordance with ASTM D3951. Adhesive shall be prepared to insure acceptance by carrier and afford protection against deterioration and damage (see 4.6.1).

5.3 Marking. In addition to any special marking required, packages and shipping containers shall be marked in accordance with MIL-STD-129 (see 4.6.1).

5.3.1 Special marking. In addition to marking as required in 5.3, containers for each part shall be properly identified. Application instructions, including cure time, number of coats required, and any special surface preparation requirements shall be marked on one part containers and on at least one of each set of two part containers. Date of manufacture and the manufacturer's batch or lot number shall be included. Container size limitations, specified in 3.11, apply.

6. NOTES

6.1 Intended use. The adhesive is intended for use where it is advantageous to use an air curing adhesive for bonding silicone rubber to aluminum and to itself. For example, to bond silicone rubber moldings to aircraft where it is not possible to use heat and pressure to facilitate a cured bond.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents should specify, but not be limited to the following:

a. Title, number, and date of this specification.

b. Military part number (see 1.2).

c. Quantity.

d. Whether a first article sample is required (see 3.1, 4.4, and 6.3).

e. Place of delivery.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of a quantity of adhesive sufficient to perform all tests specified herein. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely upon such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Adhesive
Silicone Rubber

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 8040-A126)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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